

Supplemental Material: LASEC: A Localized Approach to Service Composition in Pervasive Computing Environments

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APPENDIX A: RELATED WORK

To the best of our knowledge, thus far there is no research work that achieves the same objectives as ours. However, many related studies do exist.

Different kinds of environments require different designs of service composition mechanisms. Many existing works exist for web services which are a method for utilizing services through world wide web [19]. In these works, service registry is an important component that allows service requestors to find service providers which previously published their information to it. Multimedia streaming is area that produced a number of distributed service composition approaches [13-17]. In distributed multimedia and stream processing domain, some multimedia content may be divisible into independently routable components, e.g. audio and video flows. As a result media content adaptation services may be linked in serial, parallel and hybrid configurations to form a directed, acyclic graph of composed services [18]. There are also works on service provisioning [22] and service composition [23] in opportunistic networks, which exploit an opportunistic contact between two mobile devices when they are within communication range of each other.

When designing mechanisms for service composition in PvCE, issues of scalability and dynamicity must be taken into consideration. Both, environment and users are impacted by the scale and dynamicity of PvCE. In the area of service composition in pervasive computing many works [1-5] propose centralized solutions. Works on automatic service composition in recent environments, such cloud computing, also often base their work on centralized architectures [21]. This category of works depends on the existence of a centralized directory of services. Namely, they rely on one or more central entities to maintain the global service information in

order to make composition decisions. Service requestors submit requirements to the directory and the directory makes a decision on the list of services that should be returned to the requestors. This approach works well for environment that are well managed and relatively stable. However, the assumption on centralized control becomes impractical in scenarios with dynamic arrivals and departures of service providers, which requires frequent updates of the central entities, resulting in large system overhead. Moreover, relying on central entities to maintain global knowledge leads to inability of the system to serve the request when the central manager is compromised.

In order to address deficiencies of centralized approaches, some works exploiting distributed approaches have already been proposed. As the first step towards decentralization of service composition they introduce distributed directory of services. For example, in [9] a hierarchical directory has been proposed. The resource-poor devices depend on resource-rich devices to support service discovery and composition. However, this approach is not fully decentralized. There exist nodes that perform the task of service discovery and composition for other nodes.

Next category of distributed service composition approaches removes the need for a service directory and provides a fully distributed search for needed services. In [7] a hierarchical task-graph based approach to do service composition in ad hoc environments has been proposed. A composite service is represented as a task-graph and sub trees of the graph are computed in a distributed manner. This approach assumes that service requestor is one of the services and relies on this node to coordinate service composition. The coordinator uses global search across the whole network to do composition. The same domain of the problem was studied in [8]. It is different from [7] in terms of the way of electing coordinator. For each composite request, a coordinator is selected from within a set of nodes. The service requestor delegates the responsibility of composition to the elected

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coordinator.

The common feature of existing distributed service composition approaches is that while they perform discovery of services in the distributed way, they still require coordinator to perform service composition. However the support for the UIO-based service composition in large scale and dynamic pervasive environments is inadequate. A large number of portable and embedded devices in these environments work collaboratively in a purely ad hoc manner and without any central control. This necessitates a localized interaction model which will achieve scalable service composition minimizing both message costs and time response. To achieve this goal, we prefer a coordinator-less based service composition solution to a coordinator based one as the former appears to be more scalable than the latter. Even using dynamic coordinators (new coordinator for each request) leads to expensive message broadcasts which is not practical for pervasive computing environments with large number of dynamically changing devices.

There are also works proposed for efficient composition meeting specified QoS requirements. For example, [13, 14, 16, 20] proposed a QoS-aware service composition framework. While [16] focus is on QoS assurance and service composition is achieved in centralized way, [13 and 14] proposed a fully decentralized service composition framework. It is constructed as a service overlay network layered on top of existing Internet infrastructure. In fact, this makes this approach not feasible to implement in pervasive computing environments, especially in real-time environments: network communication has no temporal guarantees; there is no defined communication with the OS; and its QoS monitoring proposal is based on sending periodical probe frames that could be a source of nondeterminism. Besides, the execution time of their online composition algorithm is not bounded. In [20] authors further differentiate between the QoS of services themselves and the network and they propose a network-aware approach that handles the QoS of services and the QoS of the network independently.

Works in service composition share common feature, which is a globalized nature of algorithms. In a globalized algorithm, at least one node needs to maintain global information about the current task and about network status. This is unsuitable for highly dynamic and large scale environments. Gathering such global information creates huge communication overhead, especially with frequent changes of topology. Moreover, global knowledge is not always necessary, since in service composition for PvCE, it is beneficial when the distance between the services used in composition is as small as possible. To remedy the problem of maintaining global information, we have opted to adopt a localized approach, which naturally tends to select compositions with high degree of composition locality. We have learned that localized approach can significantly decrease the communication cost as well as response delay. However, we are unaware

of any localized algorithms for service composition in pervasive computing environments.

APPENDIX B: EXAMPLE OF LOCALIZED SERVICE COMPOSITION

A detailed example for G_c and G_s is shown in Fig. 1. From Fig. 1 (a), G_s is a tree graph and $V_s = \{A, B, C, D, E, F, G, H, I\}$. In G_c in Fig. 1(b)(c), the alphabet letter on each node denotes the service type provided by the node (i.e., UIO). For simplicity, we use empty circles to denote the nodes which have no corresponding node in G_s . In Fig. 1(b), the nodes make compositions just with their neighbors. After the composition process, one composition result meets the request, while others fail. However, if the nodes search other nodes in their 2-hops distance to collaborate for the composition, more composition results will meet the request, as shown in Fig. 1(c).

APPENDIX C: NP-COMPLETENESS PROOF

The SCP problem is NP-complete and we prove its NP-completeness by showing that it is a generalization of the sub-graph isomorphism problem (SGI), a NP-complete problem, which is defined as follows:

Instance: Graphs $G = (V_1, E_1)$, $H = (V_2, E_2)$

Question: Does G contain a sub-graph isomorphic to H ?

By setting $G_c = G$, $G_s = H$, SGI can be reduced to SCP, while there still exist differences between SGI and SCP:

- G and H are undirected, unlabeled and non-attributed graphs, while G_c and G_s are directed, labeled, and attributed graphs;
- SGI aims to find a sub-graph of G , while SCP aims to find an overlay of G_c ;

For the first different point, according to [10][11], the variant of SGI for directed graphs is still NP-complete. [12] has also proved that labeled SGI problem contains unlabeled SGI problem by setting the label set size to be one, and consequently, the labeled SGI problem is still a NP-complete problem. Similarly, we can prove that non-attributed graph is an extreme case of attributed graph, where the attribute size is zero. That is to say, a directed, labeled, and attributed SGI problem is still NP-complete. Furthermore, we can show that a sub-graph is a special case of an overlay. In summary, the SCP problem is a generalization of the SGI problem, and thereby it is NP-complete.

APPENDIX D: ALGORITHM PSEUDO-CODES